Cellular Differentiation Study Guide

A2.2.13— Cell differentiation as the process for developing specialized tissues in multicellular organisms. *Students should be aware that the basis for differentiation is different patterns of gene expression often triggered by changes in the environment.*

- Outline the benefits of cell specialization in a multicellular organism.

  Cells focus on fewer tasks at one making them efficient and saving energy

  Have specialized structure and metabolism

  Improve performance fast in particular tasks

- Define differentiation.

  Development of specialized structures and functions in cells

- Describe the relationship between cell differentiation and gene expression.

  Gene expression: process which the information encoded in a gene is turned into function

  gene expression controls which genes are present (shown) making cells differentiate from stem cells

A2.2.14— Evolution of multicellularity. *Students should be aware that multicellularity has evolved repeatedly. Many fungi and eukaryotic algae and all plants and animals are multicellular. Multicellularity has the advantages of allowing larger body size and cell specialization*.

- State that multicellularity has evolved repeatedly.
- List groups of organisms that are multicellular.

  Plants, animals, some fungi + algae

- Outline the steps in the evolution of multicellularity.

  Single cells —> undifferentiated multicellularity —> differentiated multicellularity

B2.3.1— Production of unspecialized cells following fertilization and their development into specialized cells by differentiation. *Students should understand the impact of gradients on gene expression within an early-stage embryo.*

- State that a zygote is an unspecialized cell produced from fertilization.
- Outline the impact of chemical gradients on gene expression within an early-stage embryo.

  Position of cell within the embryo determined how it differentiates, gradients of signaling chemicals called Morphogens impact gene expression, resulting in differentiation of the cell

  Morphogen (signaling cells) give concentration, closest to have high concentration, not too far has lower concentration to one closer differentiation, farthest has low concentration than to both differentiation

B2.3.2— Properties of stem cells. *Limit to the capacity of cells to divide endlessly and differentiate along different pathways.*

- Outline two properties of stem cells.

  Divide indefinitely, no limit to number of times stem cells can divide

  Differentiate to become specialized cell type in organism that is multicellular

B2.3.3— Location and function of stem cell niches in adult humans. *Limit to two example locations and the understanding that the stem cell niche can maintain the cells or promote their proliferation and differentiation. Bone marrow and hair follicles are suitable examples.*

- Define stem cell niche.

  The precise location of stem cells within a tissue

- Outline the location and function of two types of multipotent stem cells in an adult human body

  Location: bone marrow (hematopoietic stem cell niche)

  Function: provides protection and physical, chemical signals necessary for cell to differentiate into different blood cells

  Location: hair follicles (stem cell niche)

  Function: provides protection to stem cell and physical, chemical signals necessary for cells to differentiate into different types of skin cells

B2.3.4— Differences between totipotent, pluripotent and multipotent stem cells. *Students should appreciate that cells in early-stage animal embryos are totipotent but soon become pluripotent, whereas stem cells in adult tissue such as bone marrow are multipotent.*

- Define totipotent, pluripotent and multipotent.

  Totipotent: can form all cell types including extra-embryotic tissues

  Pluripotent: can form all body cell types but not extra-embryotic tissues

  Multipotent: “different types of blood cells, but not other cell types” (can form a limited range of related cell types

- List an example of a totipotent, pluripotent, and multipotent stem cell.

  Totipotent: zygote, embryonic

  Pluripotent: inner mass of a blastocyst(forms after fertilization, 2 parts, inner cell mass + trophoblast layer)

  Multipotent: adult stem cells, brain, bone marrow, heart, epidermis, liver, adipose tissue, skeletal muscle

- Explain why pluripotent stem cells are most prevalent in the early embryonic development of a multicellular organism.

  Early stage embryos are entirely composed of stem cells after an egg is fertilized

B2.3.5— Cell size as an aspect of specialization. *Consider the range of cell size in humans including male and female gametes, red and white blood cells, neurons and striated muscle fibers.*

- Relate the relative cell size to the specialized function of sperm, egg, red blood cell, white blood cell, neuron and striated muscle fibers.

  Largest to smallest: Egg cell (110 micro),

  Striated Muscle Fibre cell (10cm+ or 100,000+ micro, narrow 20-100 micro)

  Motor Neuron cell (cell body: 20 micro, axon can extend: 1,000,000+ micro)

  White blood cells (active: 30 micro)

  Human Sperm cell (50 micro)

  White Blood cell (inactive: 10 micro)

  Red Blood cell (width: 6-8 micro, thickness 1 micro)

C3.1.2— Cells, tissues, organs and body systems as a hierarchy of subsystems that are integrated in a multicellular living organism. *Students should appreciate that this integration is responsible for emergent properties. For example, a cheetah becomes an effective predator by integration of its body systems.*

- Define tissue, organ and organ systems.

  Tissues: cells united to preform specific function (can have two types of cells)

  Organs: tissues united to work together to carry out specific function, tissues within an organ are interdependent

  Organ systems: group of organs interact with each other to perform an overall funtion of life

- Outline how integration occurs between and among tissues, organs and organ systems.

  The organs of animal bodies are integrated to form a functioning whole. This integration involves effective communication and the transport of materials and energy.

  Tissue level integration (cell-to-cell communication), organ level integration (multiple tissues working together), organ system level integration (interdependence of organs), whole body integration (communication across organ systems)

- Define emergent property.

At each level of biological organization, new properties or behaviors emerge that are not present in the lower levels

When parts interact to become a whole new property

- State an example of an emergent property for each level of biological organization within a multicellular organism.

Characteristics of life emerge at the cellular level of biological organization,

heart tissues are able to have synchronized contractions due to interaction amongst cardiac cells,

collectively all of the tissues of the heart are able to work together to pump blood, together the components of the cardiovascular system are able to transport blood throughout the body,

organism is able to use the blood to perform all of the interconnected functions needed to survive and reproduce

C2.2.16- Consciousness as a property that emerges from the interaction of individual neurons in the brain. *Emergent properties such as consciousness are another example of the consequences of interaction.*

- Define emergent property.

At each level of biological organization, new properties or behaviors emerge that are not present in the lower levels

- State that new properties emerge at each level of biological organization.
- Outline consciousness as an emergent property.

Collective interactions of neurons in a brain

A neuron does not have a conscious thought on its own but multiple do (brain)